

Design and Analysis of A Single-Stage Transistor Amplifier Using C++

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To design a transistor based audio amplifier, many formulae are required to calculate the various parameters such as values of resistors, coupling capacitor, bypass capacitor, power consumption, current flowing through different components and others. A good amount of time is required to calculate such parameters even with a regular calculator. Using C++ program, all the parameters can be calculated in a fraction of a second.

To design an electronic circuit like a power supply, an amplifier, oscillator or operational amplifier, certain formulae are required so that correct values of components are obtained for the intended application. Some circuits require only one formula, while others may require two or more.

The single-stage transistor audio amplifier presented here requires a total of 19 formulae to obtain the values of components and analyse the circuit.

Single-stage transistor amplifier

The single-stage transistor amplifier consists of a single npn transistor T1 (BC547), four resistors R1, R2, Rc and Re, two coupling capacitors C1 and C2, one bypass capacitor Ce connected across emitter resistor Re as shown in Fig. 1.

+Vcc is the supply voltage, Vi the input signal and Vo the output signal. I1, I2, Ic and Ie, as shown in the figure, are DC currents. Resistors R1, R2 and Re are used to provide proper voltages to base b, emitter e and collector c of T1, so that base-

emitter junction is forward-biased and base-collector junction is reverse-biased.

Collector resistor Rc is used as the load that provides output voltage Vo. Bypass capacitor Ce is used to bypass the AC audio-frequency signal towards the ground so that constant

$$V_e = 0.1 V_{cc}$$

If voltage across the collector and emitter is Vce = 0.5Vcc, the remaining voltage of 0.4Vcc appears across the collector resistance.

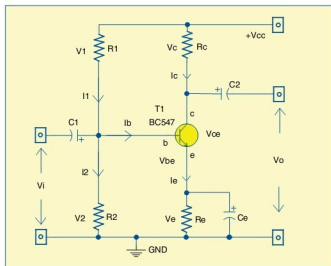


Fig. 1: Circuit diagram of the single-stage transistor amplifier

DC emitter voltage Ve is maintained to prevent negative feedback in the amplifier. Coupling capacitors C1 and C2 are used to block the DC signal and pass the AC signal.

Design guidelines

In order to bias the transistor properly with voltage-divider bias method, operating point Q should be at the centre of the DC load line (not shown here), where Vce = Vcc/2 and collector current Ic is as per the data sheet of the transistor. With no AC input signal, stable operating Q is maintained by using the one-tenth rule, which makes emitter voltage one-tenth of the supply voltage, Vcc. Therefore emitter voltage in volts is given by:

sented by β or h-parameter hFE).

Therefore emitter current in milliamperes is given by:

$$I_e = I_c + I_b$$

Emitter resistance of the transistor in kilo-ohms is given by:

$$R_e = V_e / I_e$$

Applying 10:1 rule of the firm voltage divider, second base resistance in kilo-ohms is given by:

$$R_2 = 0.1 \times \beta \times R_e$$

The base voltage or voltage across R2 in volts is given by:

$$V_2 = V_e + V_{be}$$

where Vbe is the voltage across the base and emitter.

For the silicon transistor, Vbe = 0.7 volts. In this case, the silicon transistor is considered. Voltage

Therefore collector resistance in kilo-ohms is given by:

$$R_c = 4 \times R_e$$

where Re is the emitter resistance.

Base current in milliamperes is given by:

$$I_b = I_c / \beta$$

where Ic is the collector current and β is the current gain of the transistor. (It is normally repre-